



PHYS121 Integrated Science-Physics

W4T4 Equilibrium and Elasticity

References:

- [1] David Halliday, Jearl Walker, Resnick Jearl, 'Fundamentals of Physics', (Wiley, 2018)
 - [2] Doug Giancoli, 'Physics for Scientists and Engineers with modern physics', (Pearson, 2009)
 - [3] Hugh D. Young, Roger A. Freedman, 'University Physics with Modern Physics', (Pearson, 2012)
- And others specified when needed.



11.11.1. A star is rotating about an axis that passes through its center. When the star “dies,” the balance between the inward pressure due to the force of gravity and the outward pressure from nuclear processes is no longer present and the star collapses inward and its radius decreases with time. Which one of the following choices best describes what happens as the star collapses?

- a) The angular velocity of the star remains constant.
- b) The angular momentum of the star remains constant.
- c) The angular velocity of the star decreases.
- d) The angular momentum of the star decreases.
- e) Both angular momentum and angular velocity increase.



Learning Outcomes

- Analyze and solve static equilibrium problems.
- Explain the elasticity of different kinds of materials.
- Find the elastic force/stress/pressure in simple situations.

Equilibrium

- We often want objects to be stable despite forces acting on them
- Consider a book resting on a table, a puck sliding with constant velocity, a rotating ceiling fan, a rolling bicycle wheel with constant velocity
- These objects have the characteristics that:
 1. The linear momentum of the center of mass is constant
 2. The angular momentum about the center of mass, or any other point, is constant

- Such objects are in **equilibrium**

$$\vec{P} = a \text{ constant} \quad \text{and} \quad \vec{L} = a \text{ constant.} \quad \text{Equation (12-1)}$$

- In this chapter we are largely concerned with objects that are not moving at all; $P = 0$ and $L = 0$.
- These objects are in **static equilibrium**
- The only one of the examples from the previous page in static equilibrium is the book at rest on the table

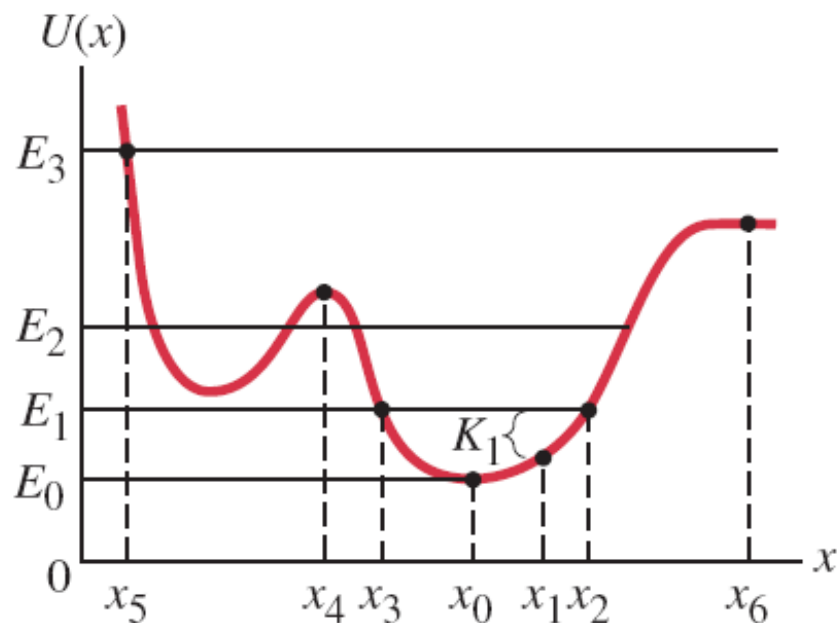
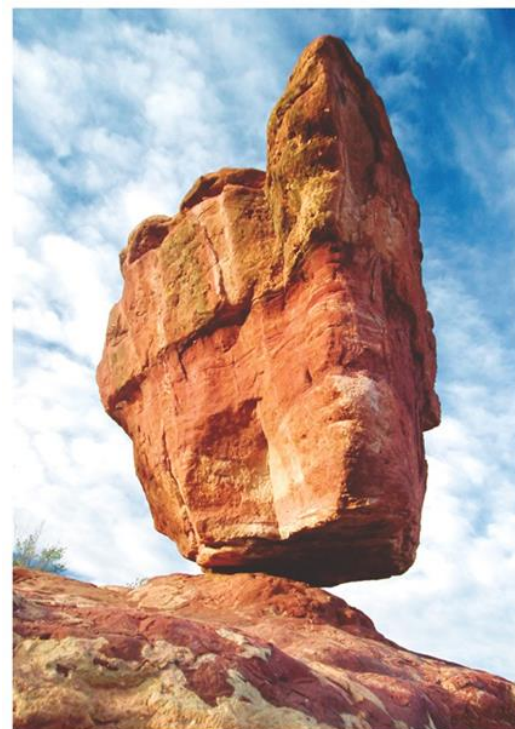


Figure 12-1



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- As discussed in 8-3, if a body returns to static equilibrium after a slight displacement, it is in stable static equilibrium
- If a small displacement ends equilibrium, it is *unstable*
- Despite appearances, this rock is in stable static equilibrium, otherwise it would topple at the slightest gust of wind

- In part (a) of the figure, we have unstable equilibrium
- A small force to the right results in (b)
- In (c) equilibrium is stable, but push the domino so it passes the position shown in (a) and it falls
- The block in (d) is even more stable

To tip the block, the center of mass must pass over the supporting edge.

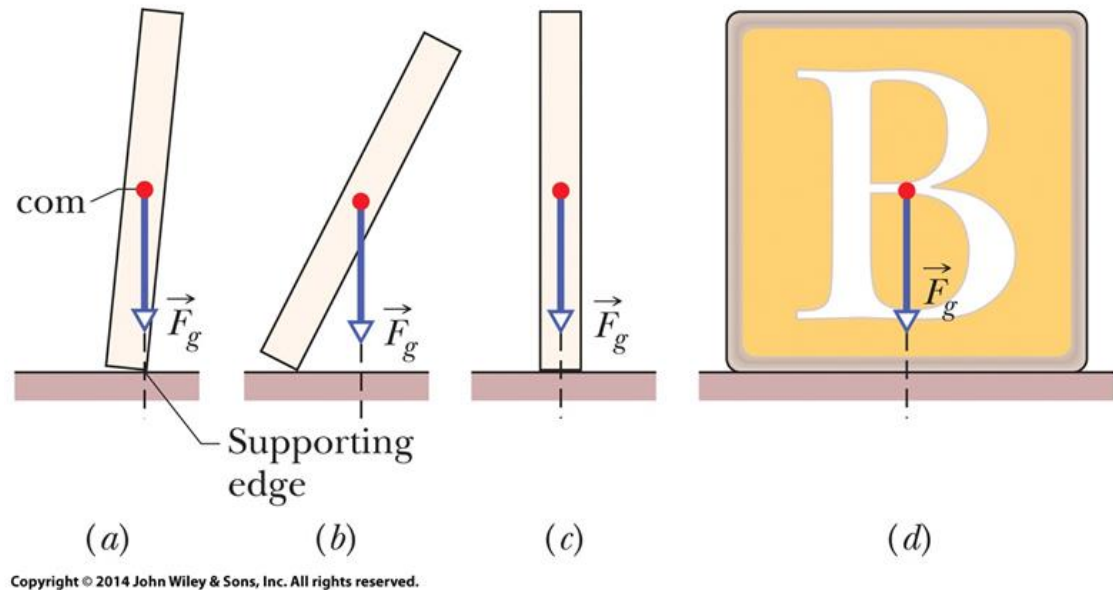


Figure 12-2

- Requirements for equilibrium are given by Newton's second law, in linear and rotational form

$$\vec{F}_{\text{net}} = 0 \quad (\text{balance of forces}). \quad \text{Equation (12-3)}$$

$$\vec{\tau}_{\text{net}} = 0 \quad (\text{balance of torques}). \quad \text{Equation (12-5)}$$

- Therefore we have for equilibrium:
 1. The vector sum of all the external forces that act on the body must be zero.
 2. The vector sum of all external torques that act on the body, measured about any possible point/axis, must also be zero.

- We often simplify matters by considering forces only in the xy plane, giving:

$$F_{\text{net}, x} = 0 \quad (\text{balance of forces}), \quad \text{Equation (12-7)}$$

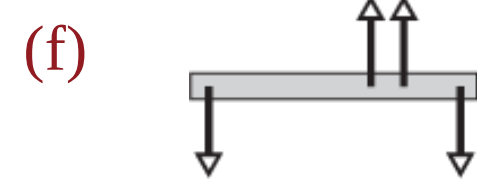
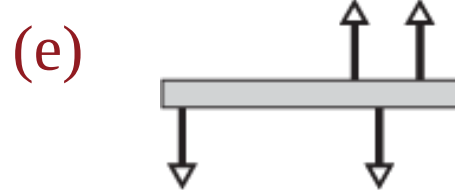
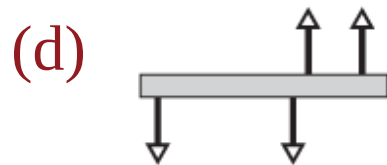
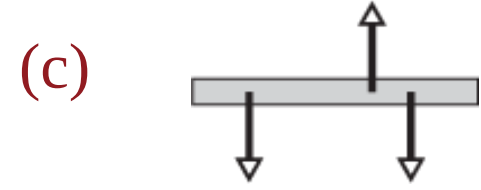
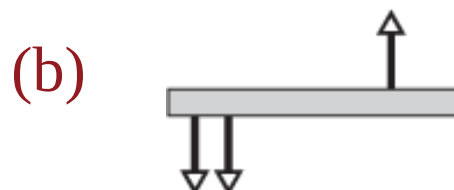
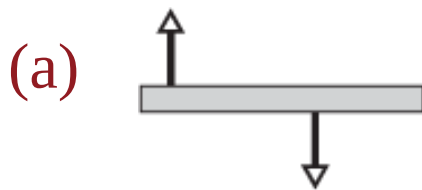
$$F_{\text{net}, y} = 0 \quad (\text{balance of forces}), \quad \text{Equation (12-8)}$$

$$\tau_{\text{net}, z} = 0 \quad (\text{balance of torques}). \quad \text{Equation (12-9)}$$

- Note that for static equilibrium we have the additional requirements that:
 3. The linear momentum $\vec{P}$ of the body must be zero.
 4. The angular momentum of the body $\mathbf{L}$ must be zero.

Checkpoint 1

The figure gives six overhead views of a uniform rod on which two or more forces act perpendicularly to the rod. If the magnitudes of the forces are adjusted properly (but kept nonzero), in which situations can the rod be in static equilibrium?



Answer:

(c), (e), (f)

- The gravitational force on a body is the sum of gravitational forces acting on individual elements (atoms) of the body
- We can simplify this by saying:

The gravitational force $\vec{F}_g$ on a body effectively acts at a single point, called the **center of gravity** (cog) of the body.

- *Until now we have assumed that the gravitational force acts at the center of mass*
- This is approximately true for the everyday case:

If $\vec{g}$ is the same for all elements of a body, then the body's center of gravity (cog) is coincident with the body's center of mass (com).

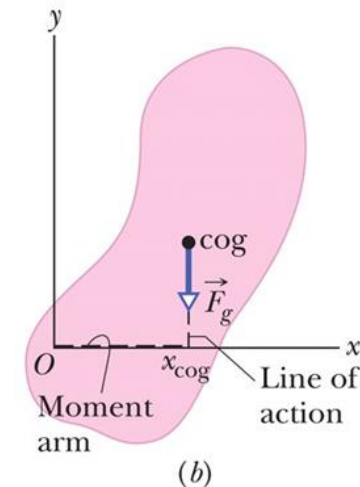
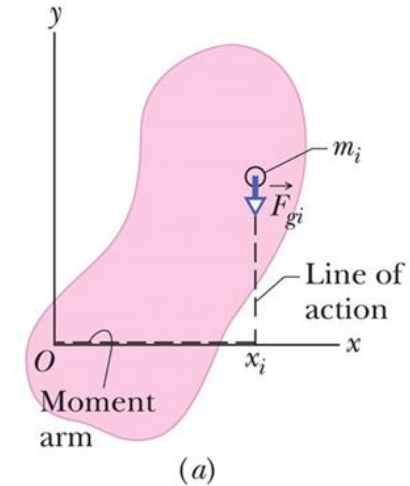
- We can show this by considering a sum of torques on each element vs. the torque caused by the gravitational force at the cog

$$x_{\text{cog}} \sum F_{gi} = \sum x_i F_{gi}.$$

- Substitute $m_i g_i$ for F_{gi}
- Cancel g ($= g_i$ for all i) and divide by the total mass, leaving:

$$x_{\text{cog}} = \frac{1}{M} \sum x_i m_i. \quad \text{Equation (12-16)}$$

- The term on the right is the com

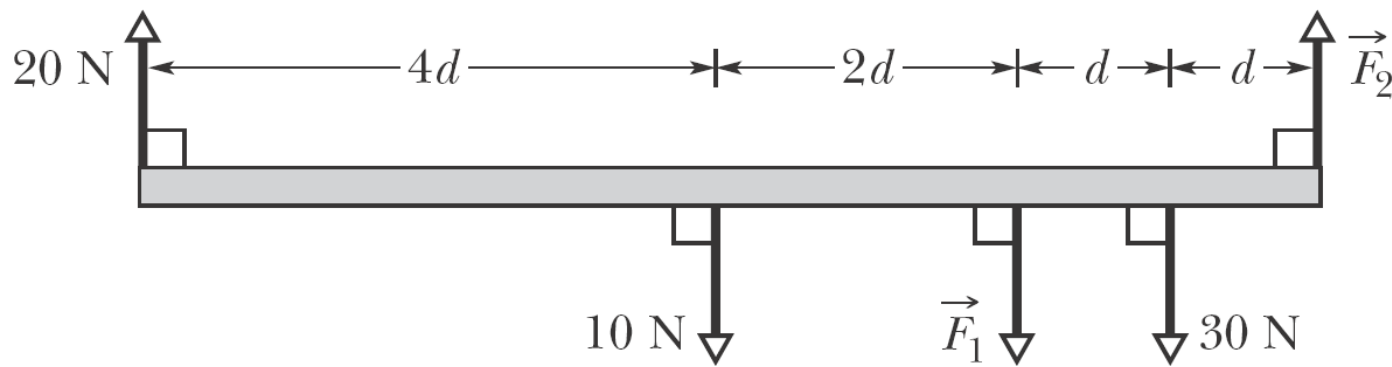


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Figure 12-4

Checkpoint 2

The figure gives an overhead view of a uniform rod in static equilibrium. (a) Can you find the magnitudes of unknown forces $\vec{F}_1$ and $\vec{F}_2$ by balancing the forces? (b) If you wish to find the magnitude of force $\vec{F}_2$ by using a balance of torques equation, where should you place a rotation axis to eliminate $\vec{F}_1$ from the equation? (c) The magnitude of $\vec{F}_2$ turns out to be 65 N. What then is the magnitude of $\vec{F}_1$?



Example Balancing a horizontal beam

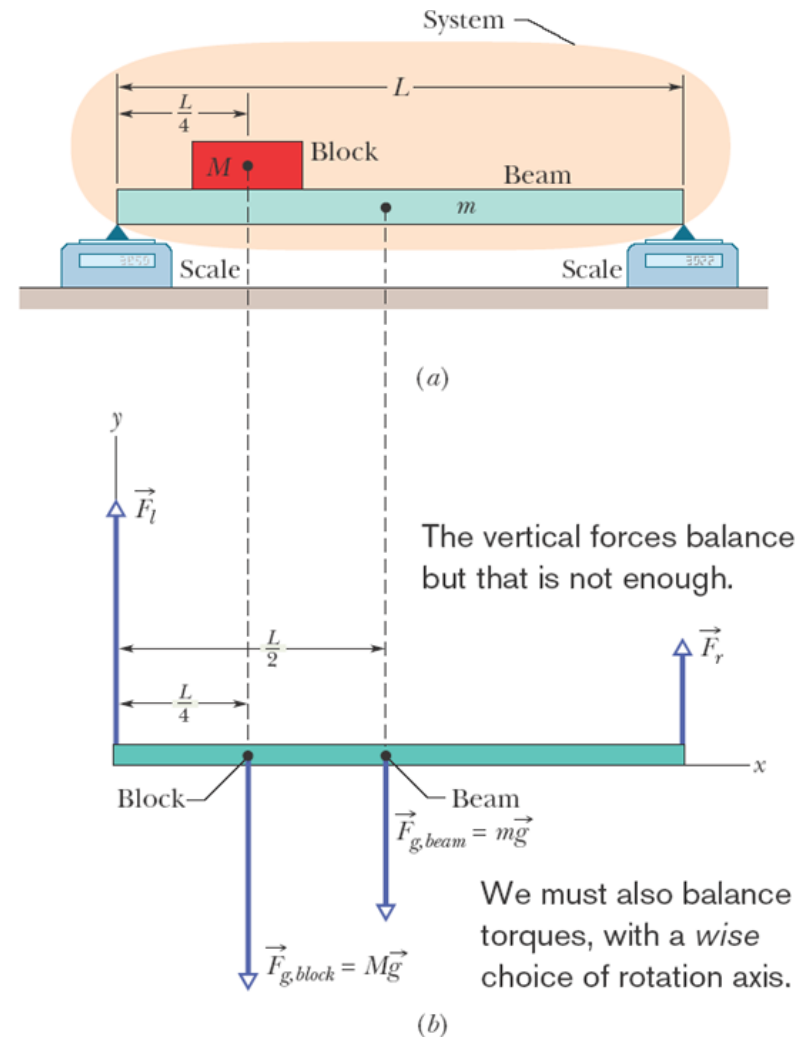
- $M = 2.7 \text{ kg}$, $m = 1.8 \text{ kg}$
- Set rotation axis at $x = 0$
- Sum torques

$$\frac{1}{4}MgL + \frac{1}{2}mgL = F_r L$$

$$\text{so } F_r = 15 \text{ N}$$

- Balance vertical forces

$$\begin{aligned} F_l &= (M + m)g - F_r \\ &= 29 \text{ N} \end{aligned}$$



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Figure 12-5

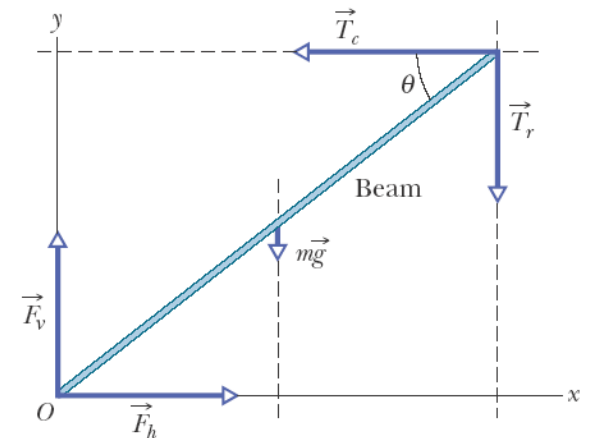
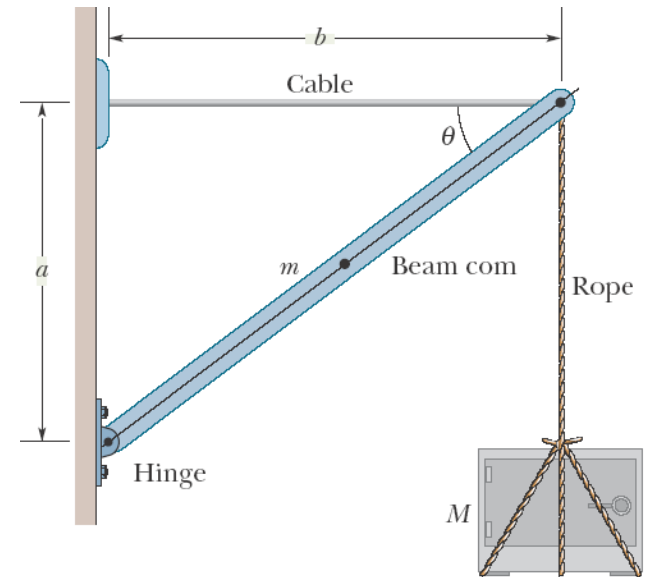
Example Balancing a leaning beam:

Find the tension in the cable, in the rope and the size of force F by the hinge.

$M = 430 \text{ kg}$, $m = 85 \text{ kg}$, $a = 1.9 \text{ m}$, $b = 2.5 \text{ m}$

- Set rotation axis at $x = 0, y = 0$
- Sum torques (using $T_r = Mg$)
- $aT_c - bT_r - \frac{1}{2}bmg = 0 \quad T_c = 6100 \text{ N}$
- Balance forces

F_h	=	$T_c = 6100 \text{ N}$
F_v	=	$(m + M)g = 5050 \text{ N}$
F	=	7900 N



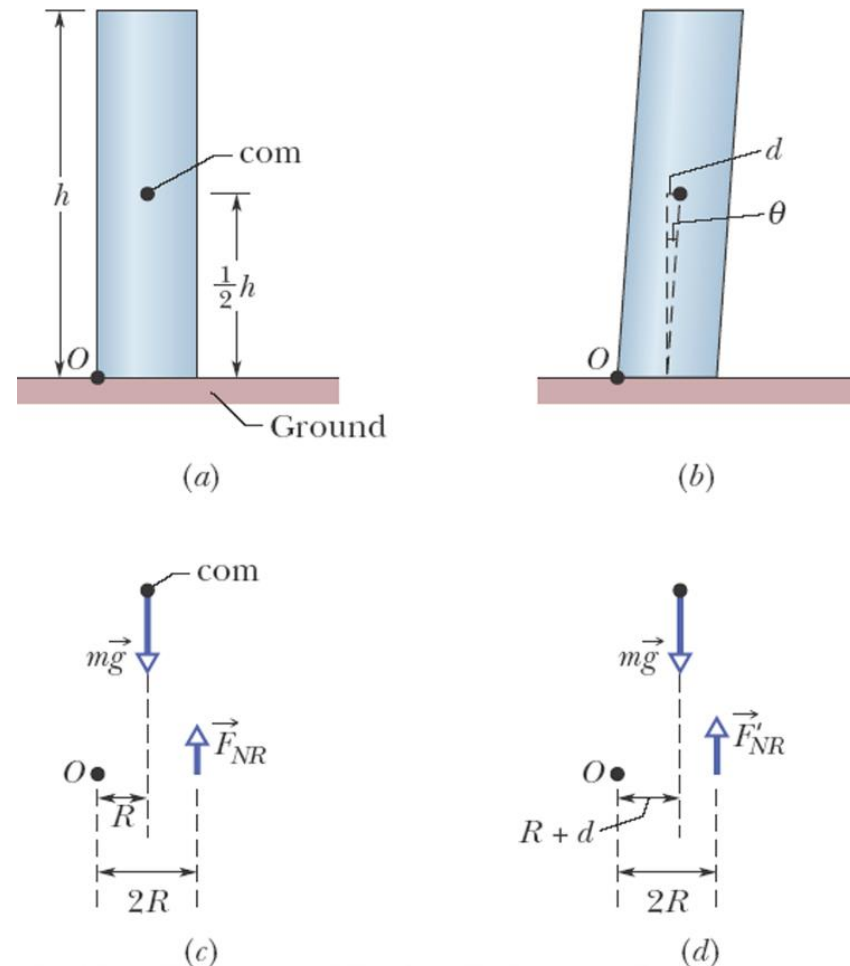
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Figure 12-6

Example Leaning Tower of Pisa

- $R = 9.8 \text{ m}$, $h = 60 \text{ m}$, $\theta = 5.5^\circ$
- Model: supported by 2 forces, at left and right edges
- No lean: $F_{NR} = \frac{1}{2}mg$
- Lean shifts com by
 - $d = \frac{1}{2}h \tan \theta$
- New force:
 - $F'_{NR} = \frac{(R + d)mg}{2R}$
- Force increases by 30%

$$\frac{F'_{NR}}{F_{NR}} = 1.29$$



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Figure 12-8

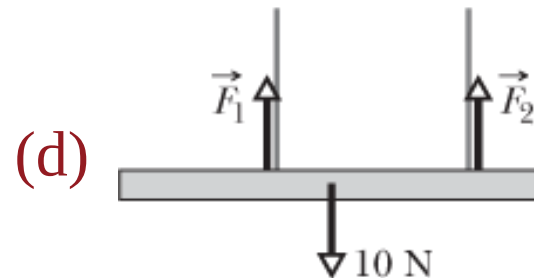
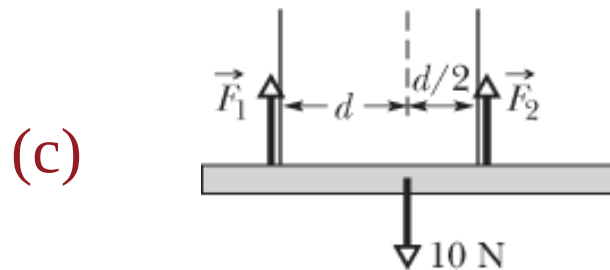
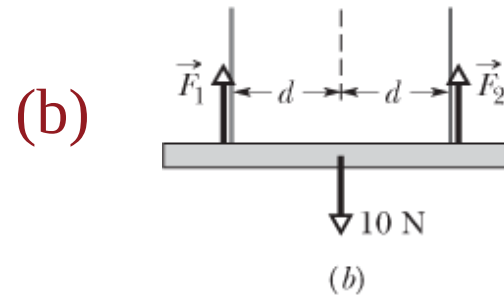
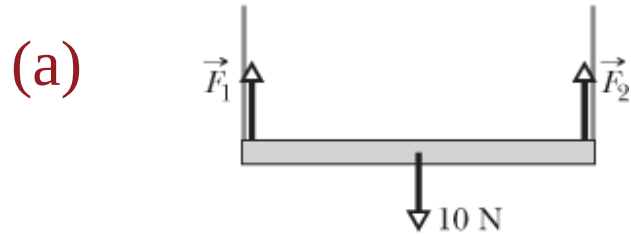
Elasticity

- For problems in the xy plane we have 3 independent equations
- Therefore we can solve for 3 unknowns
- If we have more unknown forces, we cannot solve for them and the situation is **indeterminate**
- This *assumes* that bodies are rigid and *do not deform* (there are no such bodies)
- With some knowledge of elasticity, we can solve more problems

Checkpoint 3

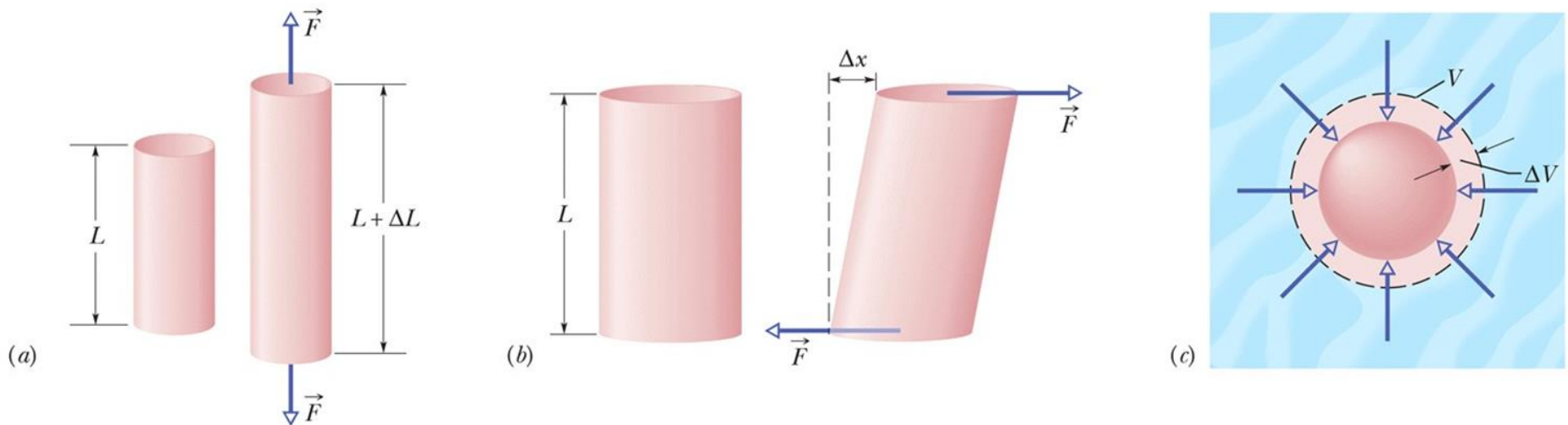
A horizontal uniform bar of weight 10 N is to hang from a ceiling by two wires that exert upward forces $\vec{F}_1$ and $\vec{F}_2$ on the bar.

The figure shows four arrangements for the wires. Which arrangements, if any, are indeterminate (so that we cannot solve for numerical values of $\vec{F}_1$ and $\vec{F}_2$)?



Answer: (d)

- All rigid bodies are partially **elastic**, meaning we can change their dimensions by applying forces
- A **stress**, *deforming force per unit area*, produces a **strain**, or *unit deformation*
- There are 3 main types of stress:
 - Tensile (a), Shearing (b), Hydraulic (c)

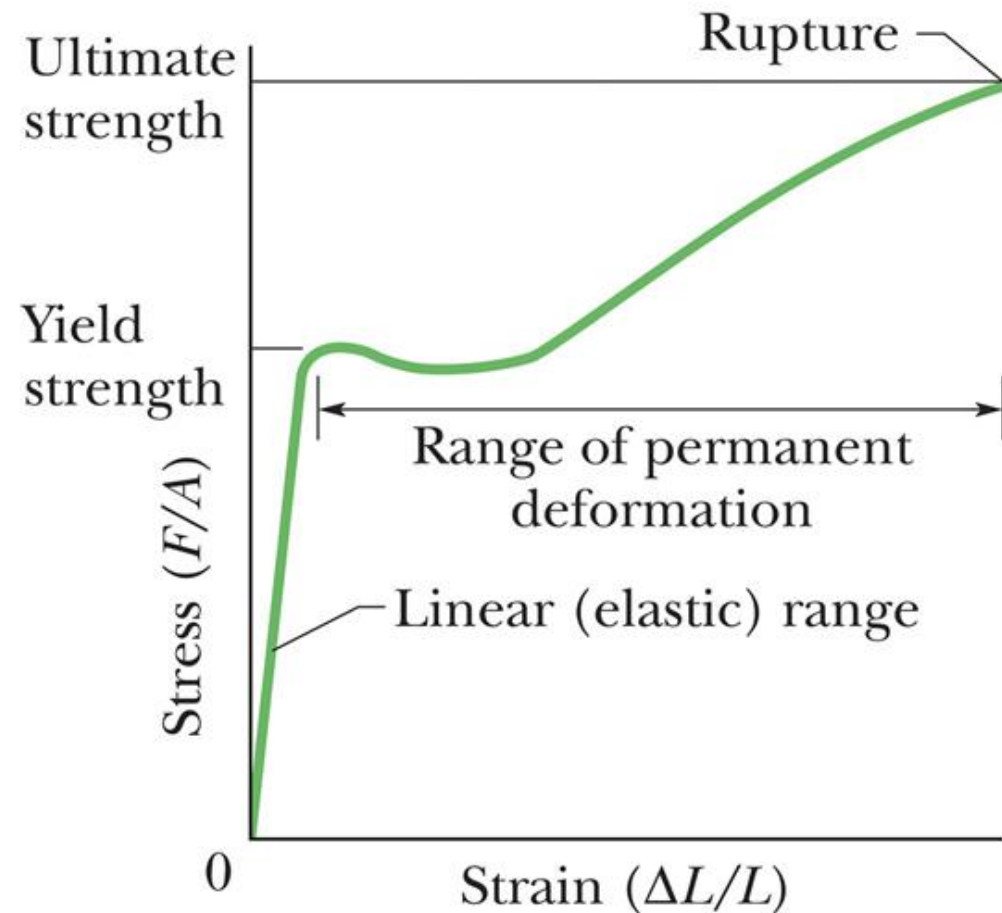


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Figure 12-11

- *Stress and strain are proportional in the elastic range*
- Related by the **modulus of elasticity**:

$$\text{stress} = \text{modulus} \times \text{strain}.$$
- As stress increases, eventually a **yield strength** is reached and the material deforms permanently
- At the **ultimate strength**, the material breaks



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Figure 12-13

- In simple tension/compression, stress is $\frac{F}{A}$
- The strain is the dimensionless quantity $\frac{\Delta L}{L}$
- **Young's modulus**, E , used for tension/compression

$$\frac{F}{A} = E \frac{\Delta L}{L}.$$

Equation (12-23)

- Note that many materials have very different tensile and compressive strengths, despite the same modulus being used for both
- E.g., concrete: high compressive strength, very low tensile strength

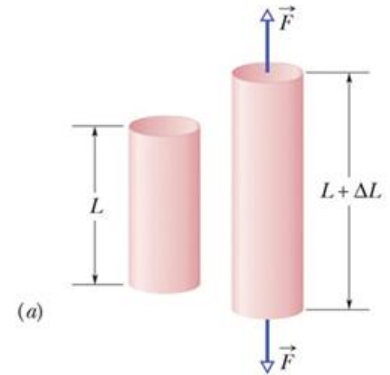
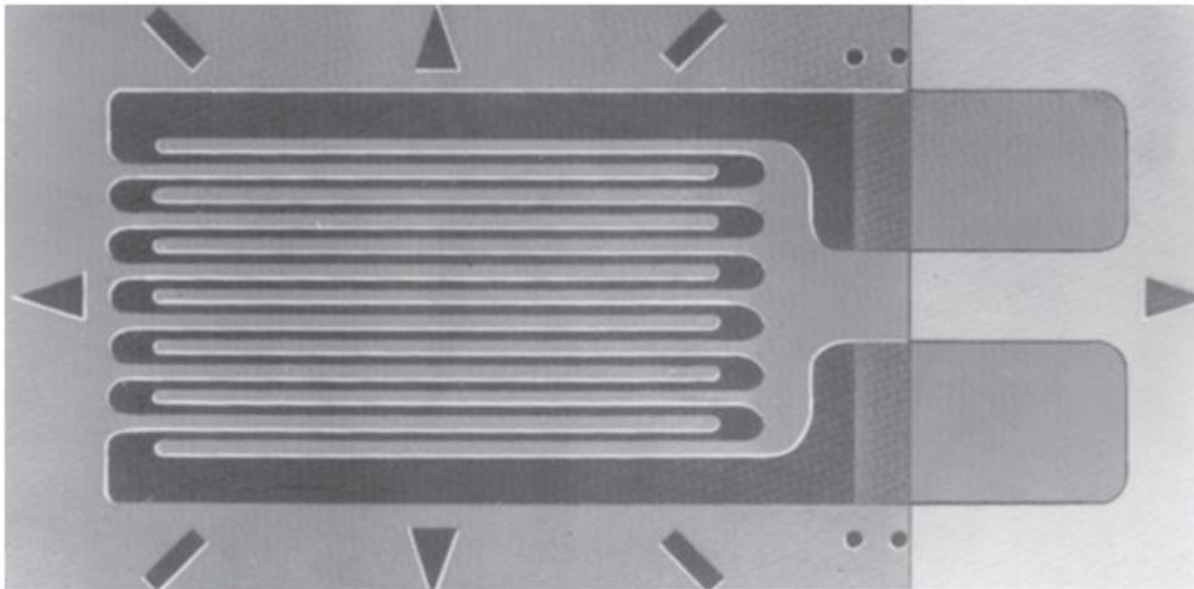


TABLE 12–2 Ultimate Strengths of Materials (force/area)

Material	Tensile Strength (N/m²)	Compressive Strength (N/m²)	Shear Strength (N/m²)
Iron, cast	170×10^6	550×10^6	170×10^6
Steel	500×10^6	500×10^6	250×10^6
Brass	250×10^6	250×10^6	200×10^6
Aluminum	200×10^6	200×10^6	200×10^6
Concrete	2×10^6	20×10^6	2×10^6
Brick		35×10^6	
Marble		80×10^6	
Granite		170×10^6	
Wood (pine) (parallel to grain) (perpendicular to grain)	40×10^6	35×10^6 10×10^6	5×10^6
Nylon	500×10^6		
Bone (limb)	130×10^6	170×10^6	

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- Strain can be measured by a strain gage
- Placed on the material, it becomes subject to the same strain
- Strain can be read out as a change in electrical resistance, for strains up to 3%

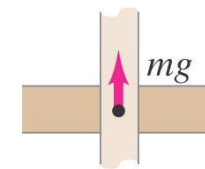
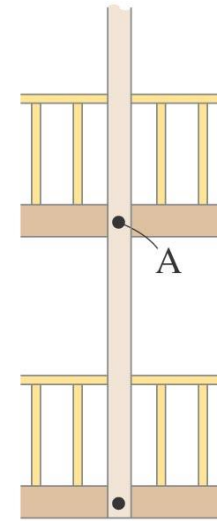


Courtesy Micro Measurements, a Division
of Vishay Precision Group, Raleigh, NC

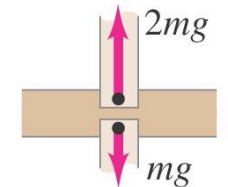
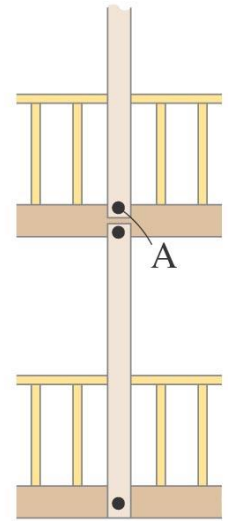
Figure 12-14

Conceptual Example: A tragic substitution.

Two walkways, one above the other, are suspended from vertical rods attached to the ceiling of a high hotel lobby. The original design called for single rods 14 m long, but when such long rods proved to be unwieldy to install, it was decided to replace each long rod with two shorter ones as shown. Determine the net force exerted by the rods on the supporting pin A (assumed to be the same size) for each design. Assume each vertical rod supports a mass m of each bridge.



Force on pin A
exerted by
vertical rod



Forces on pins
at A exerted by
vertical rods

- **Shear modulus, G** , used for shearing

$$\frac{F}{A} = G \frac{\Delta x}{L}. \quad \text{Equation (12-24)}$$

- Δx is along a different axis than L

- **Bulk modulus, B** , used for hydraulic compression

$$p = B \frac{\Delta V}{V}. \quad \text{Equation (12-25)}$$

- Relates pressure to volume change

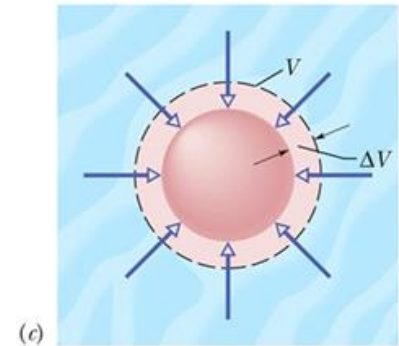
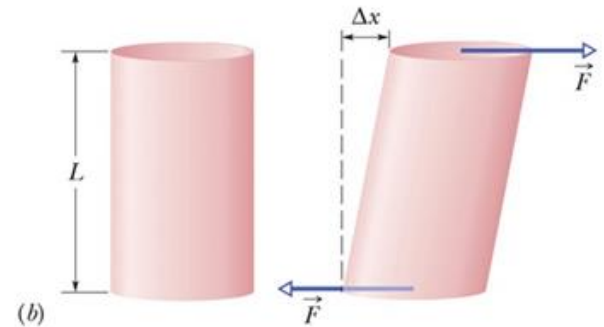


TABLE 12–1 Elastic Moduli

Material	Young's Modulus, E (N/m²)	Shear Modulus, G (N/m²)	Bulk Modulus, B (N/m²)
<i>Solids</i>			
Iron, cast	100×10^9	40×10^9	90×10^9
Steel	200×10^9	80×10^9	140×10^9
Brass	100×10^9	35×10^9	80×10^9
Aluminum	70×10^9	25×10^9	70×10^9
Concrete	20×10^9		
Brick	14×10^9		
Marble	50×10^9		70×10^9
Granite	45×10^9		45×10^9
Wood (pine) (parallel to grain)	10×10^9		
(perpendicular to grain)	1×10^9		
Nylon	5×10^9		
Bone (limb)	15×10^9	80×10^9	
<i>Liquids</i>			
Water			2.0×10^9
Alcohol (ethyl)			1.0×10^9
Mercury			2.5×10^9
<i>Gases</i> [†]			
Air, H ₂ , He, CO ₂			1.01×10^5

[†]At normal atmospheric pressure; no variation in temperature during process.

Table 12-1 Some Elastic Properties of Selected Materials of Engineering Interest

Material	Density (kg / m ³)	Young's Modulus E (10 ⁹ N / m ²)	Ultimate Strength S_u (10 ⁶ N / m ²)	Yield Strength S_y (10 ⁶ N / m ²)
Steel ^a	7860	200	400	250
Aluminum	2710	70	110	95
Glass	2190	65	50 ^b	—
Concrete ^c	2320	30	40 ^b	—
Wood ^d	525	13	50 ^b	—
Bone	1900	9 ^b	170 ^b	—
Polystyrene	1050	3	48	—

^a Structural steel (A S T M-A36). ^b In compression.

^c High strength. ^d Douglas fir.

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Example Balancing a wobbly table

- Three legs of 1.00 m, a fourth longer by 0.50 mm
- Compressed by $M = 290$ kg so all four legs are compressed but not buckled and the table does not wobble

- Legs are wooden cylinders with area $A = 1.0$ sq cm

$$\frac{F}{A} = E \frac{\Delta L}{L}.$$

- $E = 1.3 \times 10^{10}$ N/m²

- The 3 shorter legs must compress the same amount, the longer leg compresses more

- Write length comparison, use the stress-strain equation, and approximate all legs to be length L

$$\frac{F_4 L}{AE} = \frac{F_3 L}{AE} + d.$$

- Get a second equation by balancing forces

$$3F_3 + F_4 - Mg = 0,$$

- Solve the simultaneous equations to find

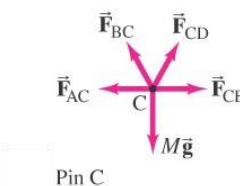
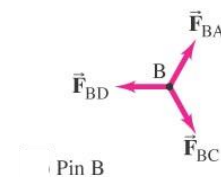
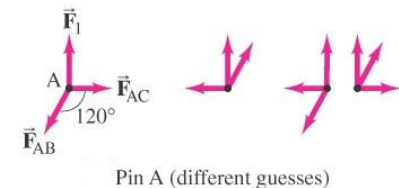
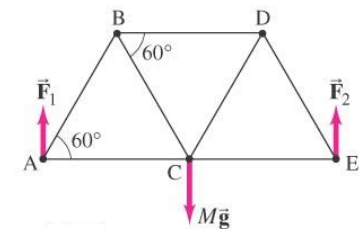
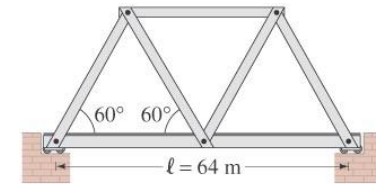
- $F_3 = 550$ N

- $F_4 = 1200$ N

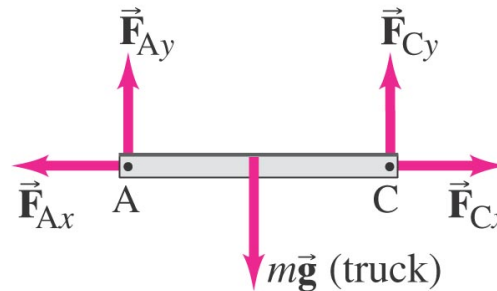
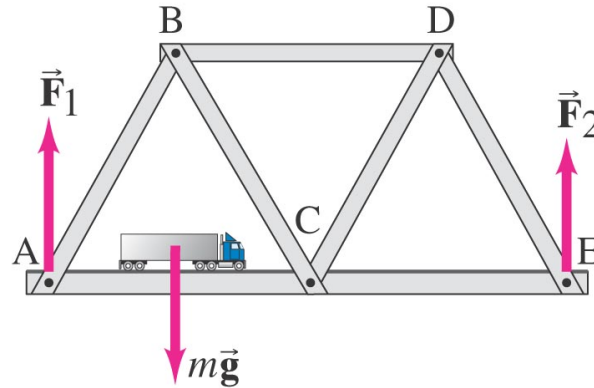
- Each short leg is compressed by 0.42 mm, and the long leg is compressed by 0.92 mm

Example: A truss bridge.

Determine the tension or compression in each of the struts of the truss bridge shown. The bridge is 64 m long and supports a uniform level concrete roadway whose total mass is 1.40×10^6 kg. Use the method of joints, which involves (1) drawing a free-body diagram of the truss as a whole, and (2) drawing a free-body diagram for each of the pins (joints), one by one, and setting $\Sigma \vec{F} = 0$ for each pin. **Ignore the mass of the struts.** Assume all triangles are equilateral.



On a real bridge, the load will not, in general, be centered. The maximum load rating for a bridge must take this into account.





Learning Outcomes

- ✓ Analyze and solve static equilibrium problems.
- ✓ Explain the elasticity of different kinds of materials.
- ✓ Find the elastic force/stress/pressure in simple situations.

$$\vec{F}_{\text{net}} = 0 \quad (\text{balance of forces}).$$

$$\vec{\tau}_{\text{net}} = 0 \quad (\text{balance of torques}).$$

$$\frac{F}{A} = E \frac{\Delta L}{L}. \quad \frac{F}{A} = G \frac{\Delta x}{L}. \quad p = B \frac{\Delta V}{V}.$$

CHAPTER REVIEW AND EXAMPLES, DEMOS

<https://openstax.org/books/university-physics-volume-1/pages/12-summary>

... static equilibrium means no motion at all and that dynamic equilibrium means motion without acceleration.

... when rotational and translational equilibrium conditions hold simultaneously in one frame of reference, then they also hold in any other inertial frame of reference, so that the net torque about any axis of rotation is still zero.

... when an object is characterized by a large value of elastic modulus, the effect of stress is small. On the other hand, a small elastic modulus means that stress produces large strain and noticeable deformation. For example, a stress on a rubber band produces larger strain (deformation) than the same stress on a steel band of the same dimensions because the elastic modulus for rubber is two orders of magnitude smaller than the elastic modulus for steel.

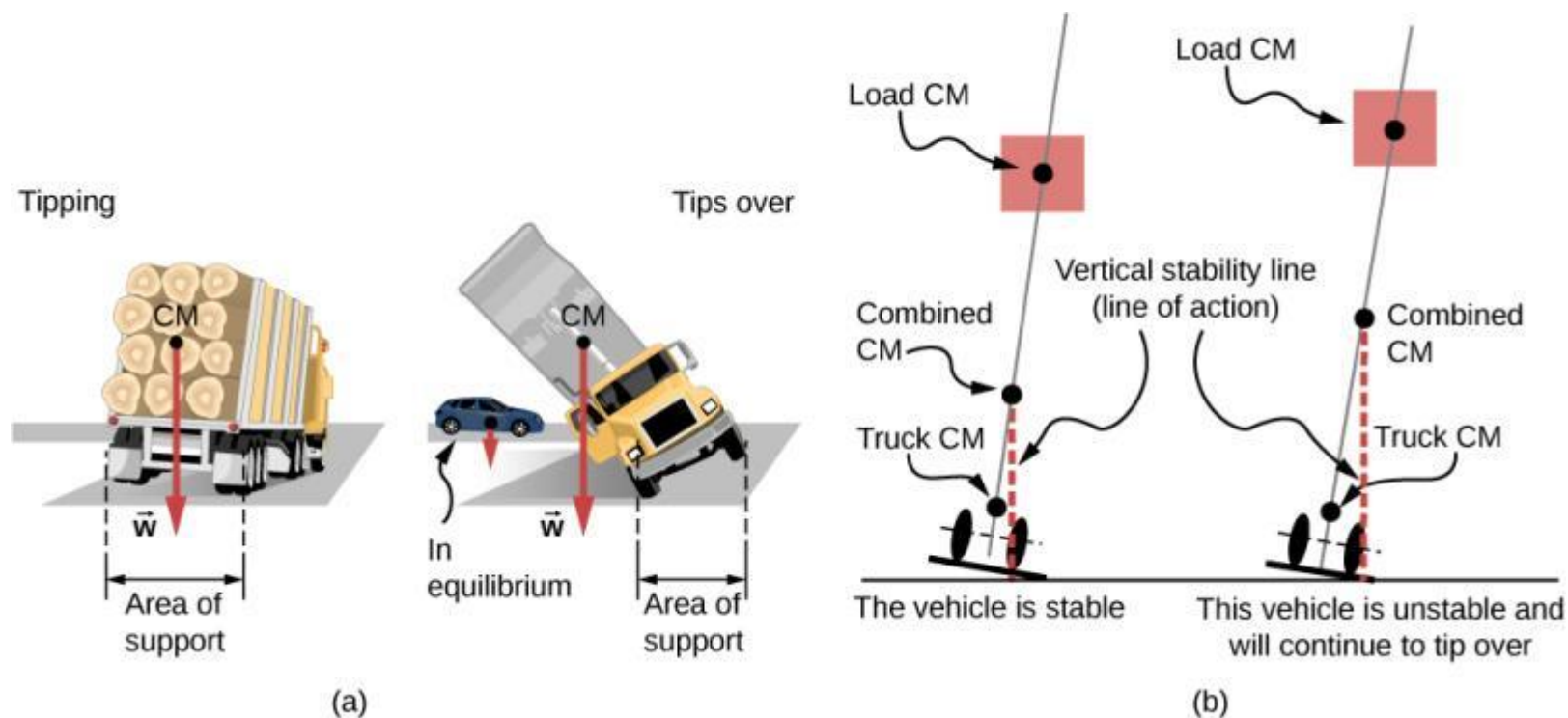
An important characteristic of pressure is that it is a scalar quantity and does not have any particular direction; that is, pressure acts equally in all possible directions.

Elasticity is the tendency of solid objects and materials to return to their original shape after the external forces (load) causing a deformation are removed. An object is **elastic** when it comes back to its original size and shape when the load is no longer present. Physical reasons for elastic behavior vary among materials and depend on the microscopic structure of the material. For example, the elasticity of polymers and rubbers is caused by stretching polymer chains under an applied force. In contrast, the elasticity of metals is caused by resizing and reshaping the crystalline cells of the lattices (which are the material structures of metals) under the action of externally applied forces.

Rubber-like materials show an increase in stress with the increasing strain, which means they become more difficult to stretch and, eventually, they reach a fracture point where they break. Ductile materials such as metals show a gradual decrease in stress with the increasing strain, which means they become easier to deform as stress-strain values approach the breaking point. Microscopic mechanisms responsible for plasticity of materials are different for different materials.

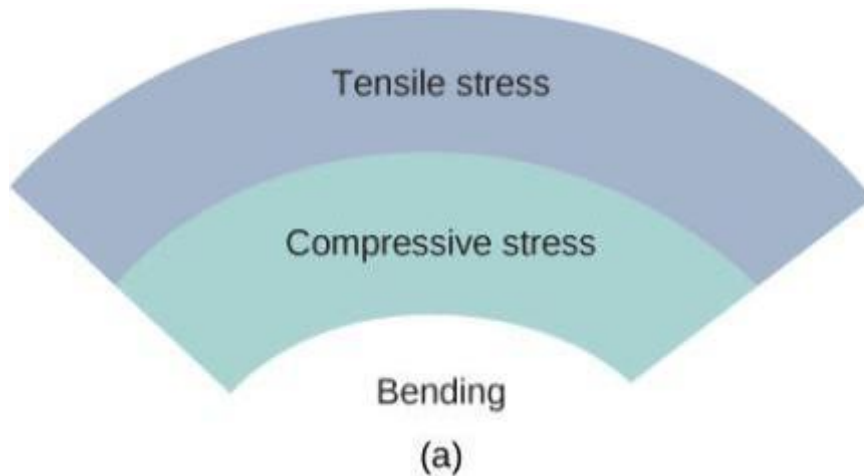
... it should be said that the linear, elastic, and plasticity limits denote a range of values rather than one sharp point.

FIGURE 12.3



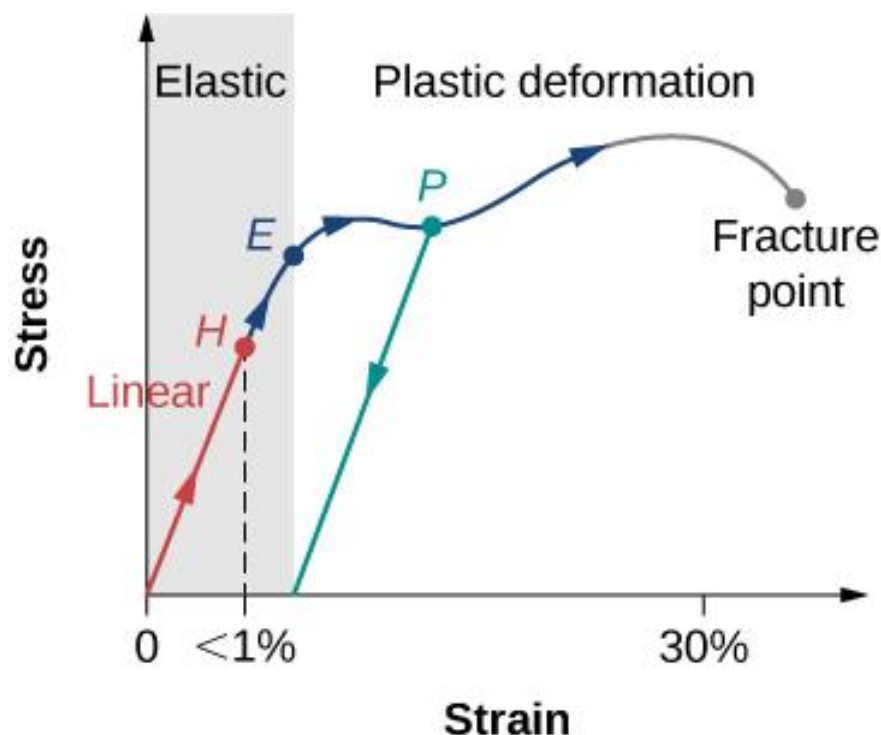
The distribution of mass affects the position of the center of mass (CM), where the weight vector $\vec{w}$ is attached. If the center of gravity is within the area of support, the truck returns to its initial position after tipping [see the left panel in (b)]. But if the center of gravity lies outside the area of support, the truck turns over [see the right panel in (b)]. Both vehicles in (b) are out of equilibrium. Notice that the car in (a) is in equilibrium: The low location of its center of gravity makes it hard to tip over.

FIGURE 12.20



- (a) An object bending downward experiences tensile stress (stretching) in the upper section and compressive stress (compressing) in the lower section.
- (b) Elite weightlifters often bend iron bars temporarily during lifting, as in the 2012 Olympics competition. (credit b: modification of work by Oleksandr Kocherzhenko)

FIGURE 12.25



Typical stress-strain plot for a metal under a load: The graph ends at the fracture point. The arrows show the direction of changes under an ever-increasing load. Points *H* and *E* are the linearity and elasticity limits, respectively. Between points *H* and *E*, the behavior is nonlinear. The green line originating at *P* illustrates the metal's response when the load is removed. The permanent deformation has a strain value at the point where the green line intercepts the horizontal axis.

Questions

